Security issues and risks

Security issues and risks associated with prompt engineering include:

Prompt injection: Users insert bad commands to take control of the AI response, overriding the intended action.

Privacy leakage and exposure: Private data may be disclosed in prompts or outputs.

Jailbreaking: This is used to circumvent a model's safeguards to create content, such as advice which is harmful or illegally restricted.

Model inversion and inference: Recovering training examples or user information from model predictions.

Resource exhaustion: An overly long or overly complex prompt that takes up extreme amounts of tokens/compute can cause a system crash.

Amplifying bias and misinformation: Prompts that present biased or false information at scale can cause real-world harm.

The art of prompt engineering is a nuanced and essential skill for maximising the effectiveness of AI models. By understanding the different types of prompts and following best practices, you can craft prompts that yield accurate and relevant results. Whether you're using zero-shot, one-shot, few-shot, or chain-of-thought prompting, the key is to be clear, provide context, and iterate based on the model's performance. With these techniques, you can harness the full potential of AI to achieve your desired outcomes. END

References

- *https://www.promptingguide.ai/*
- *https://platform.openai.com/docs/guides/text?api-mode=responses*
- *https://cloud.google.com/discover/what-is-prompt-engineering*
- *https://www.deeplearning.ai/short-courses/chatgpt-prompt-engineering-for-developers/*
- *https://www.ibm.com/think/topics/prompt-engineering-guide*
- *https://learnprompting.org/docs/introduction?srsltid=AfmBOopsyh2UUPXDiUdvw8P-ekvwr-YvriawmkAJwLUVUL9n-HaHzgQC*

By: Dr Magesh Kasthuri and Dr Anand Nayyar

Dr Magesh Kasthuri is a PhD in artificial intelligence and the genetic algorithm. He currently works as a distinguished member of the technical staff (master) and chief architect at Wipro Ltd. This article expresses his view and not of the organisation he works in.

Dr Anand Nayyar is a PhD in wireless sensor networks and swarm intelligence. He works at Duy Tan University, Vietnam, and loves to explore open source technologies, IoT, cloud computing, deep learning, and cyber security.